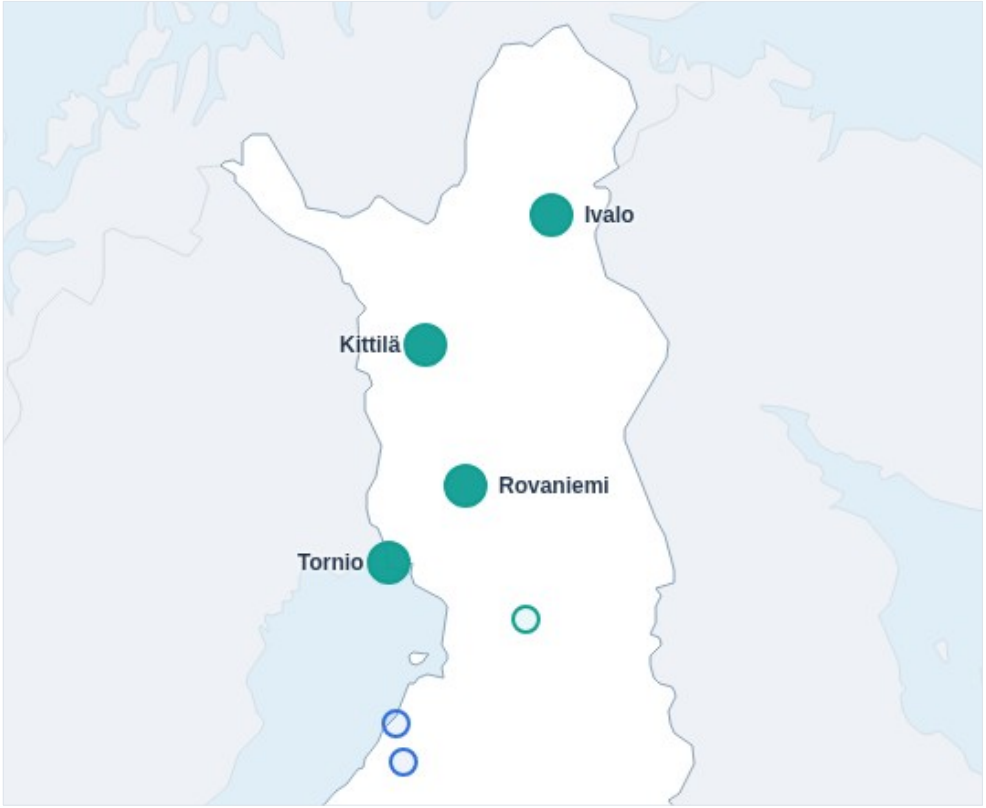
In May 2005, Kittilä and Ivalo flooded together.

The snow had warned us two months earlier.

1 March snow flagged 2005 in advance. Kittilä and Ivalo then passed their flood levels in the same spring.

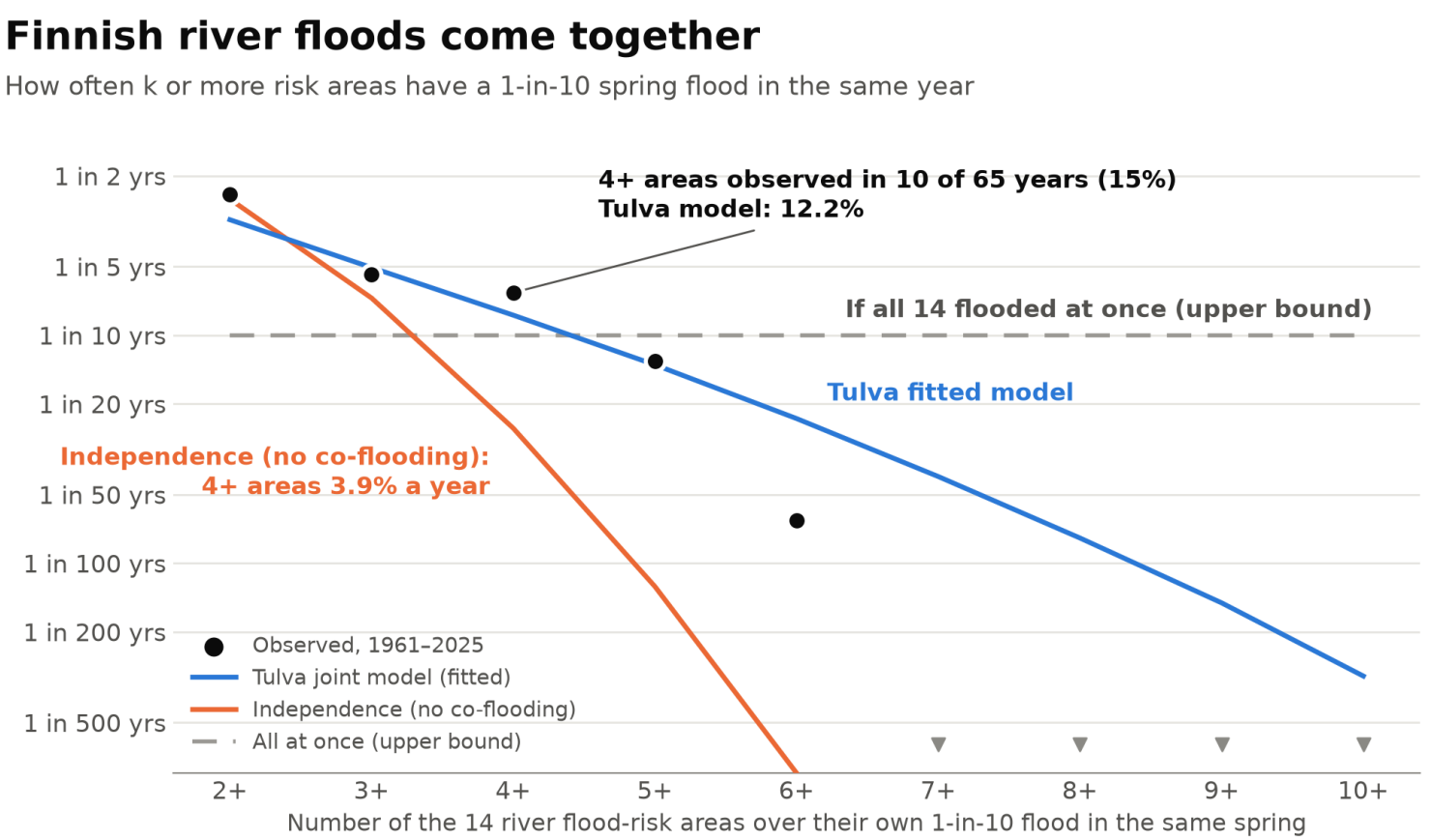


Tulva app, year replay: spring 2005

Source: SYKE discharge 1961–2025 and 1 March snow (SWE); Tulva back-test (out/backtest.json)

THE GAP

Finland's rivers flood together, not one at a time



Data: SYKE Hydrologiarajapinta (daily discharge, CC BY 4.0) and SMHI (Torne). Spring (Mar–Jun) peak flow, one gauge per area, 1961–2025.
Model: Gaussian-copula factor model (national + 3 regional factors), exact quadrature on the same gauge-years as observed. ▼ = not observed.

Springs when ≥4 of the 14 river areas flooded (own 1-in-10 level)

15.4 %

observed, 1961–2025 (10 of 65 years)

12.2 %

Tulva joint model

3.9 %

if areas were independent

Home insurance pays only for floods rarer than 1-in-50.

Reinsurance renews 1 January, before the snow signal.

Source: SYKE 1961–2025 (+ SMHI Torne), Tulva model (exact); vesi.fi / OP (1-in-50 rule)

WHAT IT COSTS

Priced as independent, a joint flood cover is ~5× too cheap

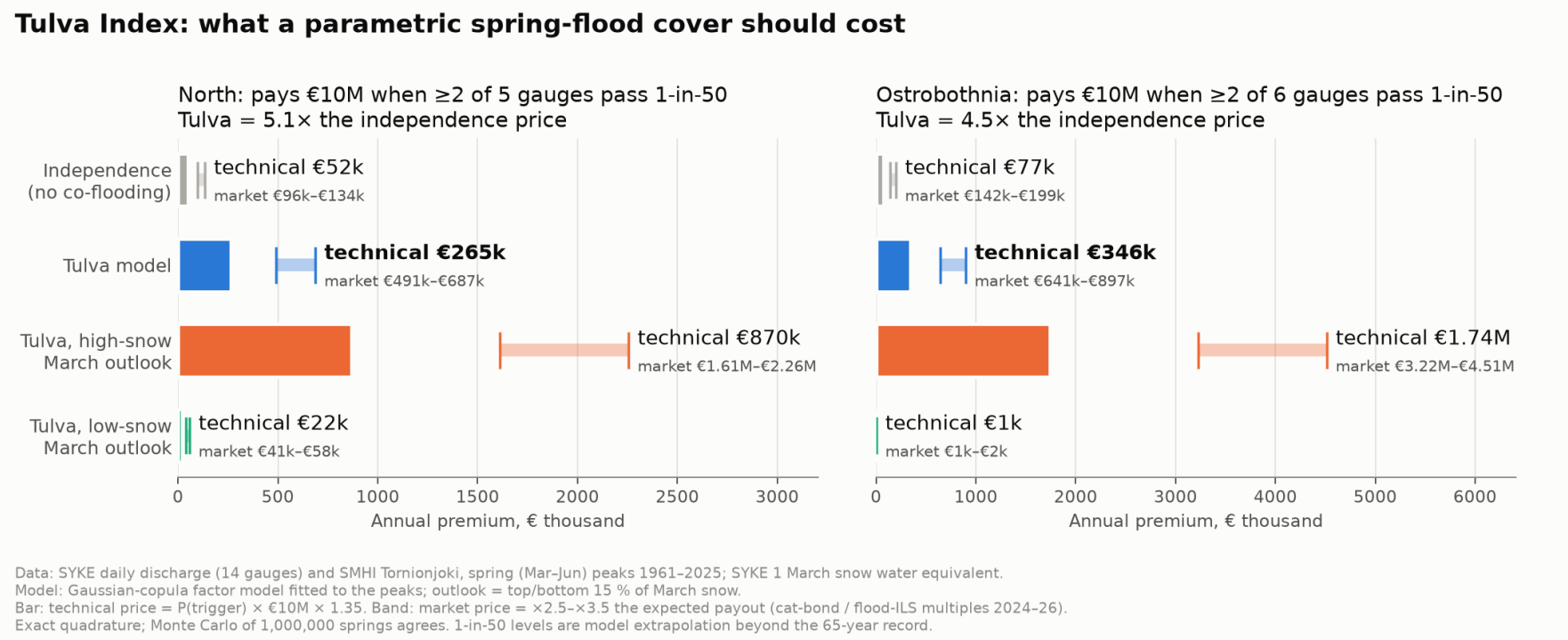
North cluster: pays €10M if ≥2 of 5 gauges pass their 1-in-50 level. Market price per year:

€96k–134k
if areas were independent

→

€491k–687k
Tulva joint risk (technical €265k)

5.1×
Tulva vs independence



Source: SYKE 1961–2025, Tulva model (exact; MC 1M agrees). Market = 2.5–3.5× expected payout (AM Best cat bonds, flood ILS)

THE PRODUCT

Tulva Index: bought in winter, pays in spring

TULVA INDEX · NORTH CLUSTER

Pays €10M if ≥2 of 5 northern gauges pass their 1-in-50 level this spring

Independence

€96k–134k

market price per year

Tulva

€491k–687k

market price per year

After a snowy March

€1.61M–2.26M

market price per year

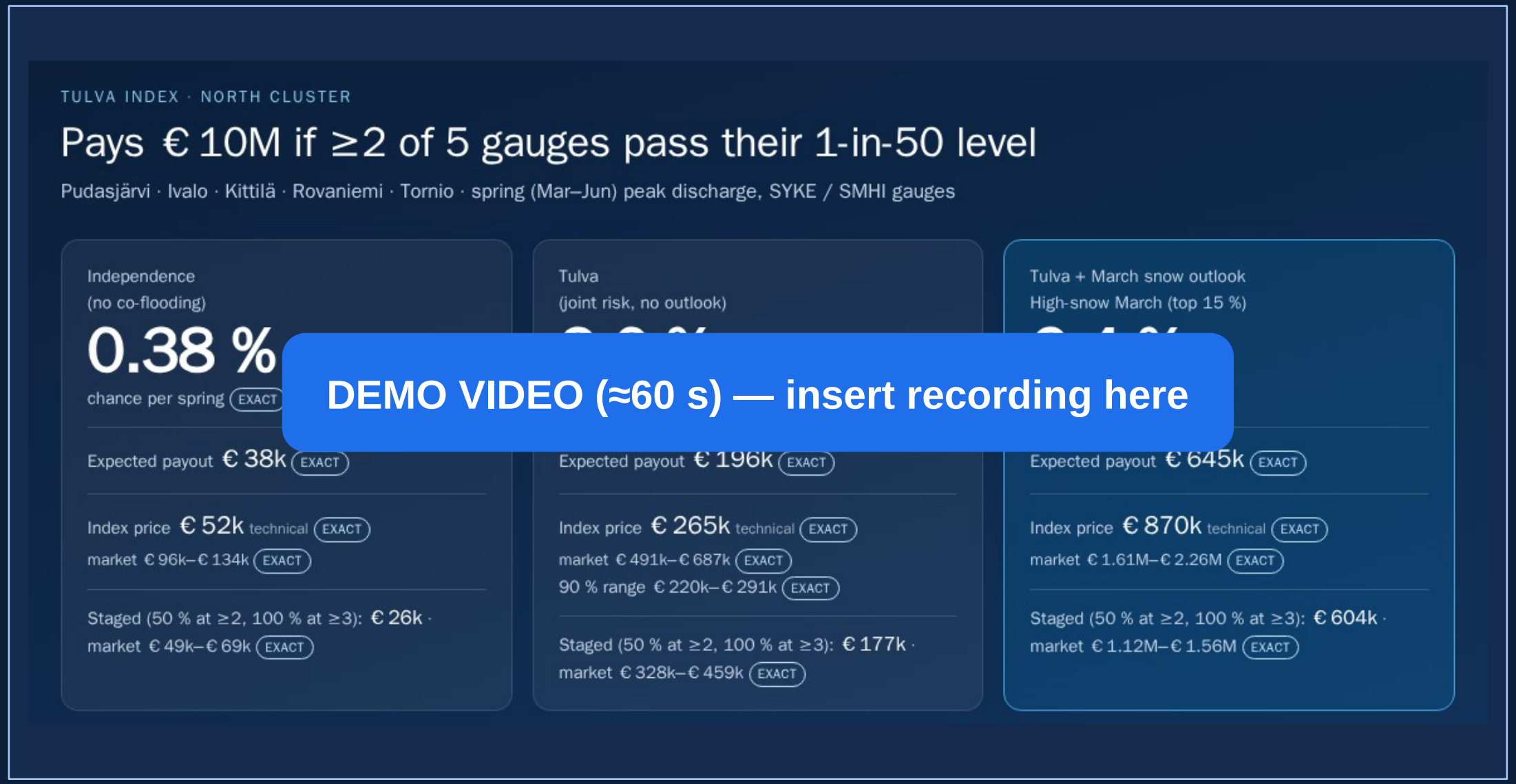
Bought in winter
before the snow signal exists

Staged option
50 % paid at 2 gauges, 100 % at 3

Explainable trigger
public SYKE gauges anyone can check

11 of 19 payout years flagged in advance by the 1 March snow, 2005 included

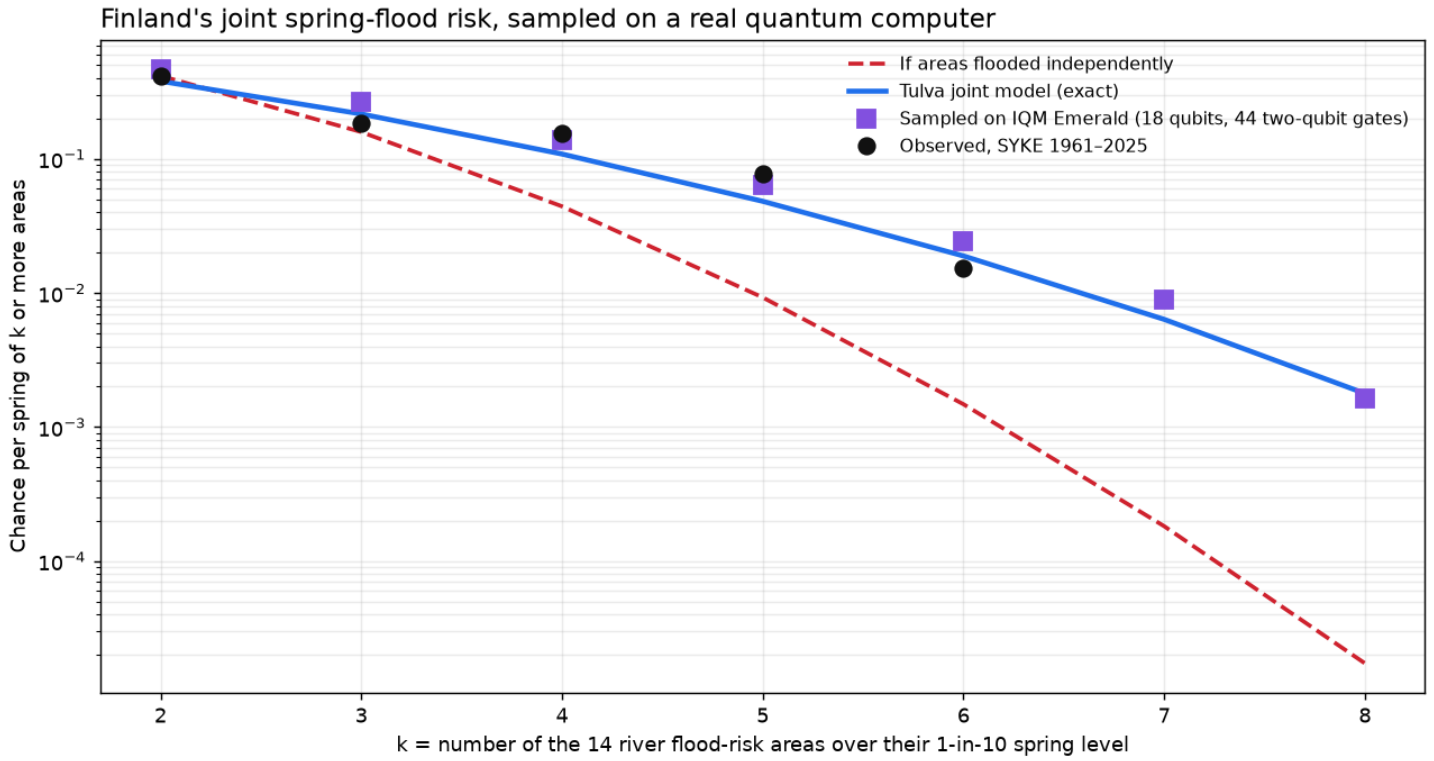
Source: Tulva model, exact; market 2.5–3.5× expected payout. Back-test 1961–2025, all clusters, ≥2 gauges over 1-in-10, in-sample



Live app: the joint-flood map → March snow outlook → index price → real quantum runs → 2005 replay

QUANTUM, IN PRACTICE

Pricing = a joint-flood probability → quantum amplitude estimation



IQM Emerald via IQM Resonance, job 01a11cd4-da38-718b-80c9-7e259c6e4452, 8,000 shots, readout-corrected. Exact = two-level factor model fitted to SYKE spring peaks.

The chip reproduces the joint tail that independence misses.

ALL 14 AREAS ON IQM EMERALD

18
qubits

44
two-qubit gates

AMPLITUDE ESTIMATION, IBM & IQM

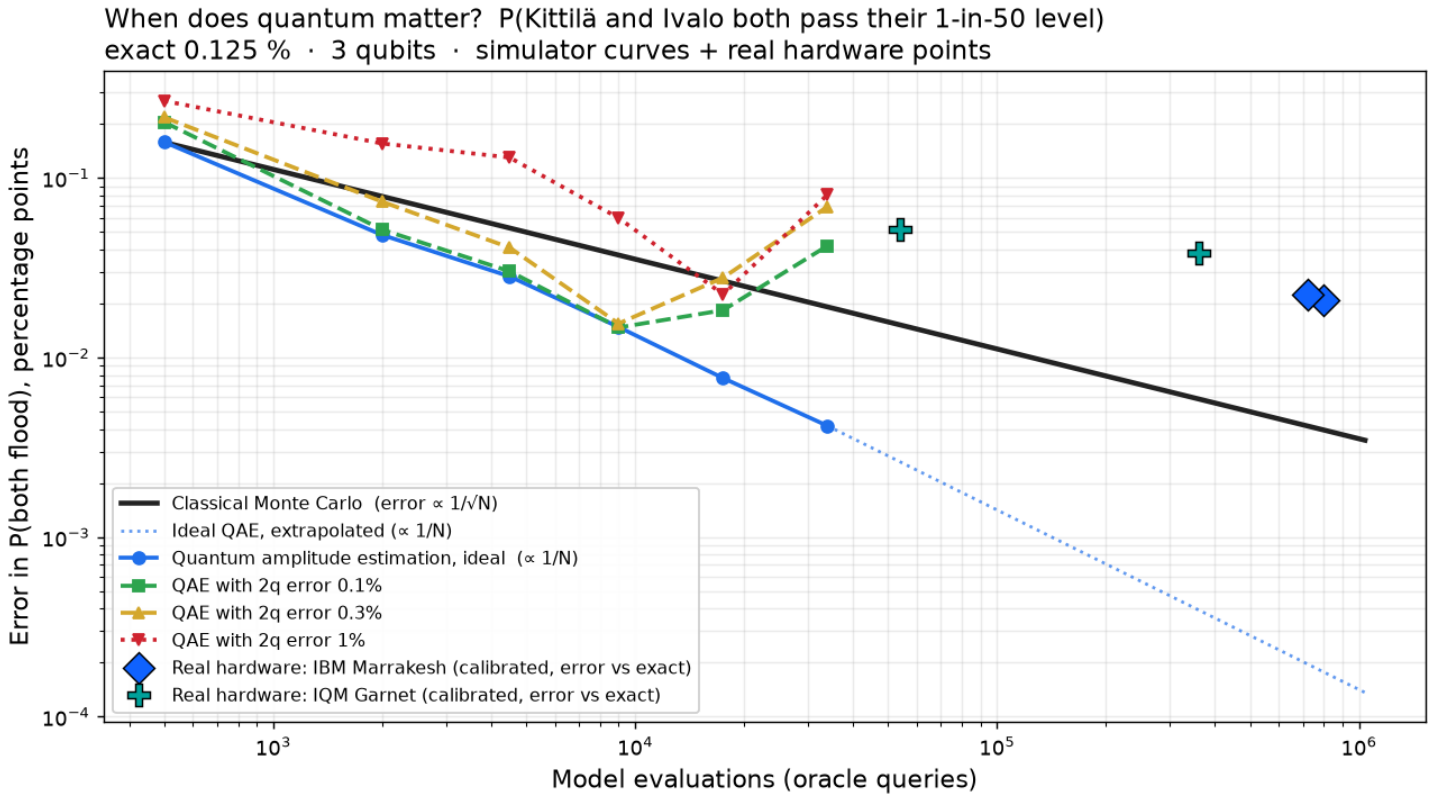
P(Kittilä and Ivalo both pass 1-in-50)

	raw → calibrated
Exact	**0.125 %**
IBM Marrakesh	0.262 → **0.148 %**
IQM Garnet	0.368 → **0.163 %**

Snow signal on the chip: **snowy vs low March 3.1× (exact 2.7×)**

IQM Emerald job 01a11cd4…, 8,000 shots; snow runs: Emerald, 10 copies, 240k shots each; IBM ibm_marrakesh (20 copies); IQM Garnet (4 copies); also native Qrisp on Garnet

Classical Monte Carlo still wins today



Fitted to SYKE spring peaks 1961–2025. Noisy curves: depolarising noise + 1 % readout, square-lattice routing; simulator error = RMS over 200 repetitions.
Hardware points = |calibrated estimate − exact| of one run (mostly residual bias), at that run's total model evaluations.

Quantum wins at scale, with:

Fault-tolerant hardware
~8k logical qubits for comparable pricing (Chakrabarti et al. 2021)

Expensive models
full hydrological simulation per scenario

More work per price
multi-peril, daily repricing

Ideal QAE error falls like 1/N, Monte Carlo like 1/√N.

Source: Tulva simulations (qae-sim) + IBM Marrakesh / IQM Garnet runs; Chakrabarti et al., Quantum 5, 463 (2021)

BUSINESS

Who buys, why, and why now

WHO BUYS

Finnish non-life insurers

For the flood they keep under reinsurance. Four groups write 91 % of the market. Then reinsurers and ILS investors.

HOW WE EARN

Engine licence

Pricing and trigger engine, licensed per insurer or reinsurer. Structuring fee on placed covers.

WHY NOW

30 Jan 2027

Solvency II flood factor for Finland: 0 % correlation with SE/NO/DK (25 % for snowmelt rejected). ORSA needs climate scenarios.

SCALE

Finland → Nordics → Central Europe

Same engine, any snowmelt or river region. Gävle 2021: SEK 1.9bn claims. Storm Boris 2024: €2.15bn insured.

First step: a paid pilot with one insurer in autumn, ahead of the 1 January renewals.

Sources: Finance Finland 2024; Del. Reg. (EU) 2026/269; EIOPA-BoS-24/462 §4.70; Finansinspektionen 2023 (Gävle); PERILS 2025 (Boris)

Tulva prices Finland's spring floods as a joint risk, and runs it on real quantum hardware.

BUILT THIS WEEKEND

- SYKE data pipeline
- Joint flood model
- Index pricing
- QAE on IBM & IQM

Toivo Kallio · **Vaishnav Rane**

Team 21 STUHI · Quantum x Finance 2026, Hanken × Ultrahack